

Multilingual Emotion Detection

Emotion detection as constrained JSON generation with Qwen3.5

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MMD II Project, Topic 3, Team 1

BRIGHTER shared task (SemEval-2025 Task 11): given a short text, predict which of six emotions it expresses.

- **Emotions:** anger, disgust, fear, joy, sadness, surprise
- **Multi-label:** any subset can be active (or none)
- **Multilingual:** 28 languages, very different scripts and resource levels

Our setup

- Train on English, Swahili, Ukrainian (8,537 examples)
- Zero-shot transfer to German, Brazilian Portuguese, Yoruba (and Chinese)

Our Approach

We treat classification as **constrained text generation** with Qwen3.5, a multilingual decoder transformer - no classification head.

- System prompt lists the six labels and the JSON schema
- User turn provides language name and text to classify
- Model emits `{"emotions": ["predicted label1", ...]}` (empty list = no emotion)
- Loss computed only on JSON completion tokens
- Language- and model-agnostic finetuning and evaluation pipeline
- Primary metric: **macro-F1** (label distribution is heavily imbalanced)

Result 1: Fine-Tuning the 0.8B Model

Trained

Language	mBERT	Base	FT 0.8B	Δ
English	0.541	0.287	0.600	+0.313
Swahili	0.238	0.176	0.273	+0.097
Ukrainian	0.400	0.300	0.501	+0.201
Mean	0.393	0.254	0.458	+0.204

Zero-shot

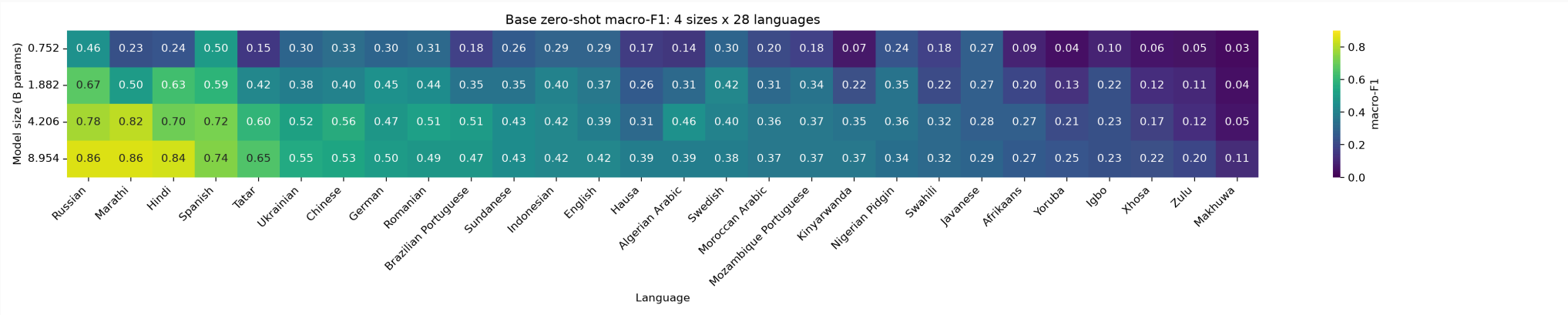
Language	mBERT	Base	FT 0.8B	Δ
German	0.310	0.302	0.391	+0.089
Brazilian Port.	0.266	0.176	0.396	+0.220
Yoruba	0.169	0.045	0.097	+0.052
Mean	0.248	0.174	0.295	+0.120

Official-dev macro-F1. Fine-tuning the 0.8B base lifts trained languages by **+0.204** and even zero-shot languages by **+0.120**.

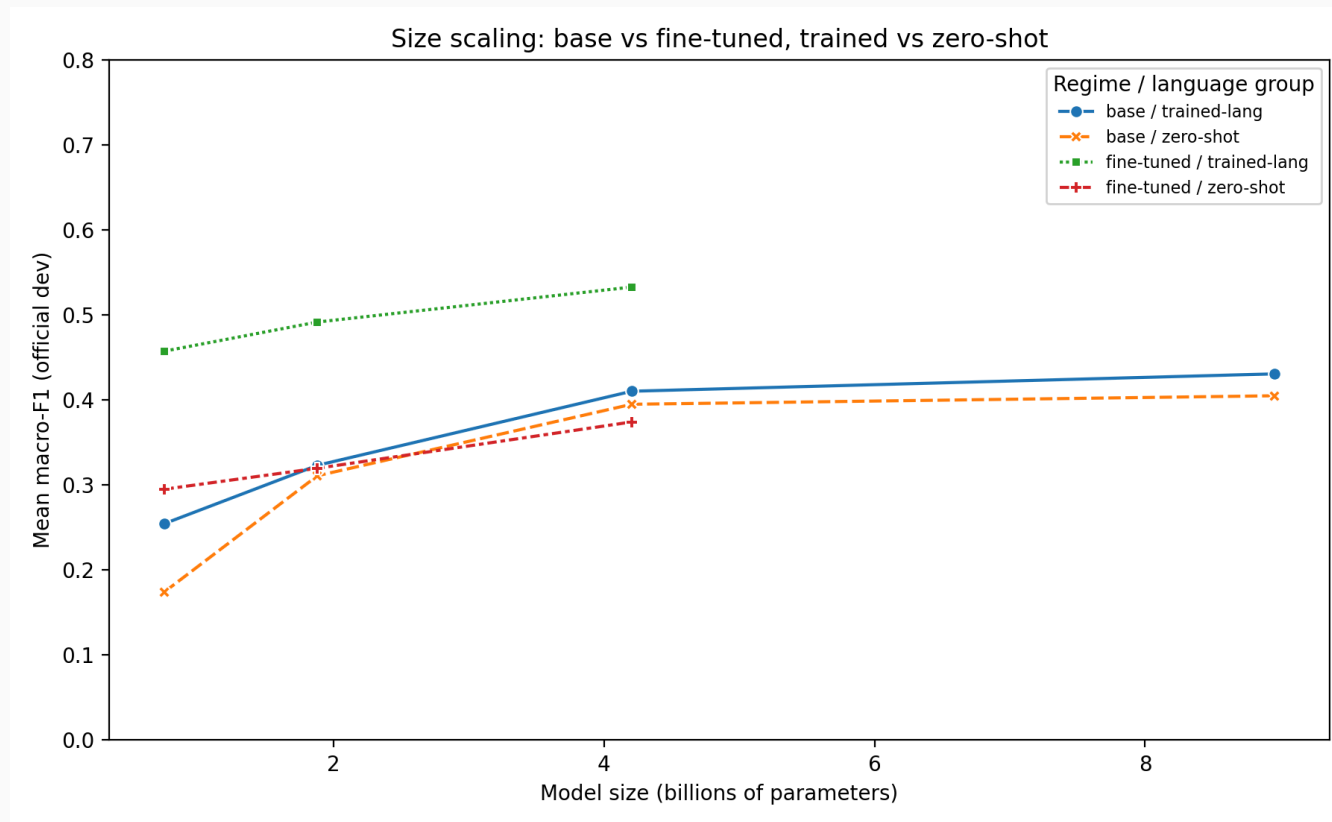
Result 2: Model Scaling Study

Size	Base tr.	Base zs.	FT tr.	FT zs.	VRAM
0.8B	0.255	0.174	0.468	0.295	1.5 GB
2B	0.323	0.311	0.495	0.328	3.8 GB
4B	0.410	0.395	0.543	0.362	8.5 GB
9B	0.431	0.405	n/a	n/a	17.9 GB

Fine-tuned **0.8B** (0.468) beats **every** larger base model, including the 9B (0.431) - 11× its size, at just 1.5 GB.



Result 2: The Scaling Trade-Off



Fine-tuning a **small** model maximises performance on known languages at minimal cost; a **large base** model maximises coverage of unknown languages.

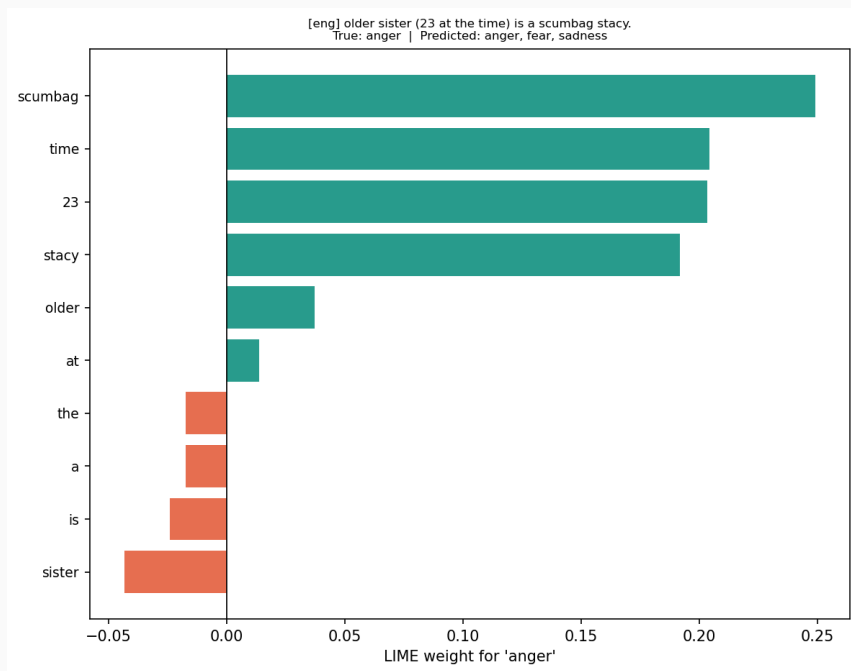
Sub-Study: Does Chinese Heritage Help?

Qwen3.5 is trained with heavy Chinese data - a natural zero-shot test case.

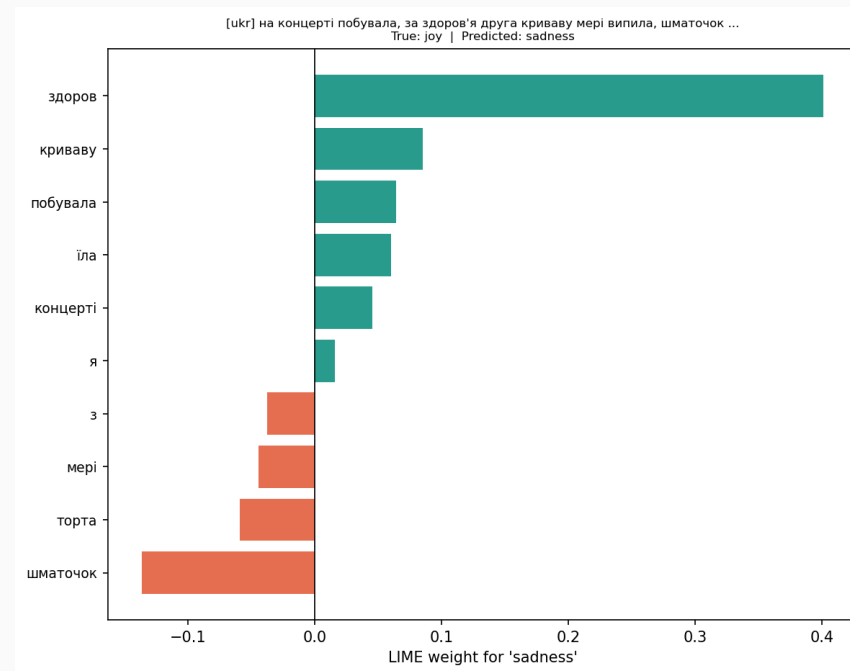
Size	Base	FT (eng+swa+ukr)	Δ
0.8B	0.332	0.466	+0.134
2B	0.399	0.483	+0.084
4B	0.563	0.555	−0.008
9B	0.531	n/a	n/a

- Strong zero-shot start: 4B base ranks 6th of 28 BRIGHTER languages
- Heritage advantage is **best preserved by a large base model**, not fine-tuning

LIME Error Analysis



English (Qwen-0.8B): “scumbag” +0.249



Ukrainian (mBERT): “здоров” +0.401 → wrong

- Explicit lexical cues transfer across scripts and model families
- Failures trace back to **pretraining coverage** (Yoruba latches on loanwords)

Two contradictory objectives

- **Known languages, minimal hardware:** 0.8B beats 9B base on trained languages
- **Unknown languages, broad coverage:** prefer a large base model

Dataset imbalances

- Labels are **heavily skewed**: macro-F1 penalises ignoring any emotion — rare emotions hurt F1 scores
- English **disgust**: zero positive training examples → F1 structurally at **0.00**
- Duplicating rare-emotion rows **hurt** (−0.023 to −0.038): more false positives, no gain

Structural limits beyond model size

- **mBERT** beats Qwen on Yoruba (0.169 vs. 0.097): it **rarely predicts** positive labels for scripts underrepresented in its pretraining data
- Chinese heritage best preserved by large base, not fine-tuning (4B > 9B without FT)

Takeaways

- A fine-tuned **0.8B** model (1.5 GB at inference) outperforms every larger base model on trained languages - including one **11× its size** → on-device viable.
- But full fine-tuning at scale **erodes zero-shot transfer**: for unseen languages, a large base model is the better choice.
- Low-resource languages (Yoruba ≈ 0.097) stay near the floor regardless of size - the bottleneck is **data**, not capacity.

The remaining lever is information-adding data, e.g. back-translation.

Thank you!
Questions?